

WATER CYCLE IN A BAG

Let's Make It Rain in a Bag!





The Mystery Puddle

Later yesterday, there was a puddle —
Where did it go today?"
Today we will catch the water cycle in a bag!
Think about it — where does the water go?

Three Magic Words

Evaporation (Steam banana) - Water goes UP!

Condensation (Drop banana) - Drops come BACK!

Precipitation / Rain - Rain falls DOWN!

Let's do the Water Cycle Chant! Say each word and do the action: Hands UP for evaporation, hands DOWN for condensation, wiggle fingers DOWN for rain!



Evaporation Explained

Sun heats water → turns to tiny invisible steam → floats UP!

Examples you see every day:

- Puddles drying after rain
- Wet clothes drying on Delhi terrace
- Steam rising from tea in a clay cup





Condensation Explained

Water vapour touches a cool surface → turns to
tiny droplets!

Examples:

- Cold glass of Lemon Water dripping outside
- Bathroom mirror fogging after bath
- Dew drops on grass in the morning



Precipitation Explained

When water droplets in clouds get heavy, they fall as rain! This is called precipitation (Rain).

- Monsoon rain in Kerala, Mumbai, Assam
 - Snowfall in Shimla or Manali
- Rivers like Ganga and Yamuna filling up

The Sun - Our Engine

The Sun gives heat energy to start the water cycle!
Without the Sun: No evaporation, no rain!
India gets strong sunshine most of the year → This brings big monsoon rains!
The Sun is the powerful engine that keeps water moving around and around!



Materials Needed

Clear zip-lock bag

Water ($\frac{1}{2}$ cup / 100 ml)

Blue food color (1-2 drops, optional)

Permanent marker

Cello tape

Sunny window

Pencil and notebook

Note: All materials cost under ₹25! Gather everything before we start our exciting water cycle experiment. Ask your teacher or parents for help!



Experiment Steps (Part 1)

- Take a clear plastic zip-lock bag
- Draw a bright sun at the top with marker
- Draw fluffy clouds in the middle

Step 4: Draw wavy water lines at the bottom of the bag. Step 5: Pour half cup (100ml) of water into the bag carefully!



Experiment Steps (Part 2)

- Add 1-2 drops blue food color
- Seal the bag tightly - no leaks!
- Tape bag to sunny window

Step 9: Write your name on the bag with marker. Step 10: OBSERVE and WAIT! Watch the magic happen over the next few hours!



What Will You See?

Tiny drops forming on sides or top?

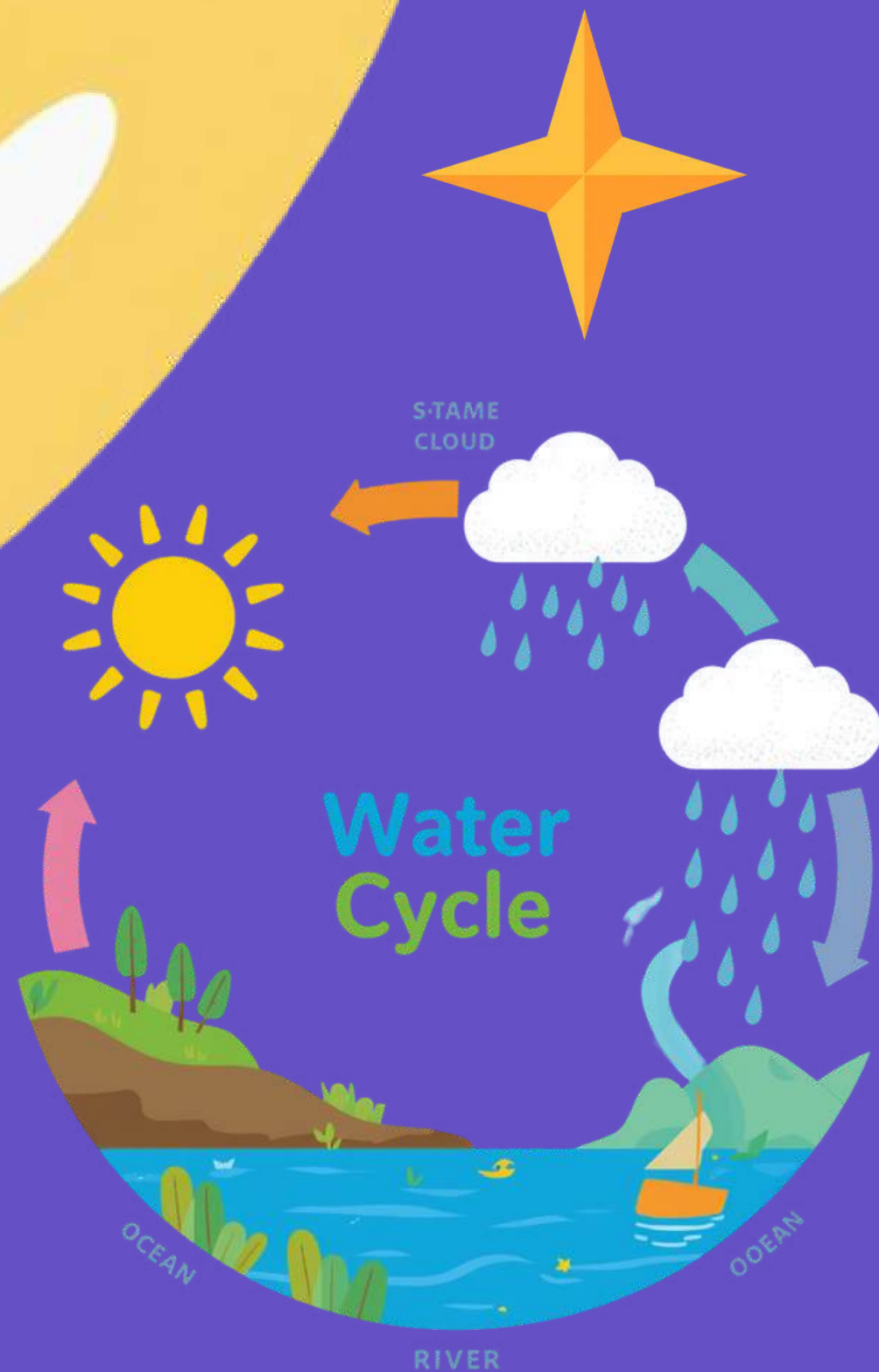
Where is the water going?

Does inside look foggy?

Drops sliding down like rain?

This is YOUR mini water cycle! Look closely at your bag and watch the magic happen. You are seeing evaporation, condensation, and precipitation — just like real rain!





The Complete Water Cycle

Watch the magic! The Sun heats water (evaporation) → Water vapour rises up → Clouds form (condensation) → Rain falls down (precipitation) → Water returns to rivers, lakes & oceans → The cycle repeats forever and ever!



India's Monsoon Connection

Indian Ocean heats in summer → lots of evaporation!
Big monsoon clouds form → heavy rain (June-Sept).
Rivers like Ganga fill up → farmers get water for crops. Our bag experiment shows the same cycle!

Key Vocabulary

Water Cycle

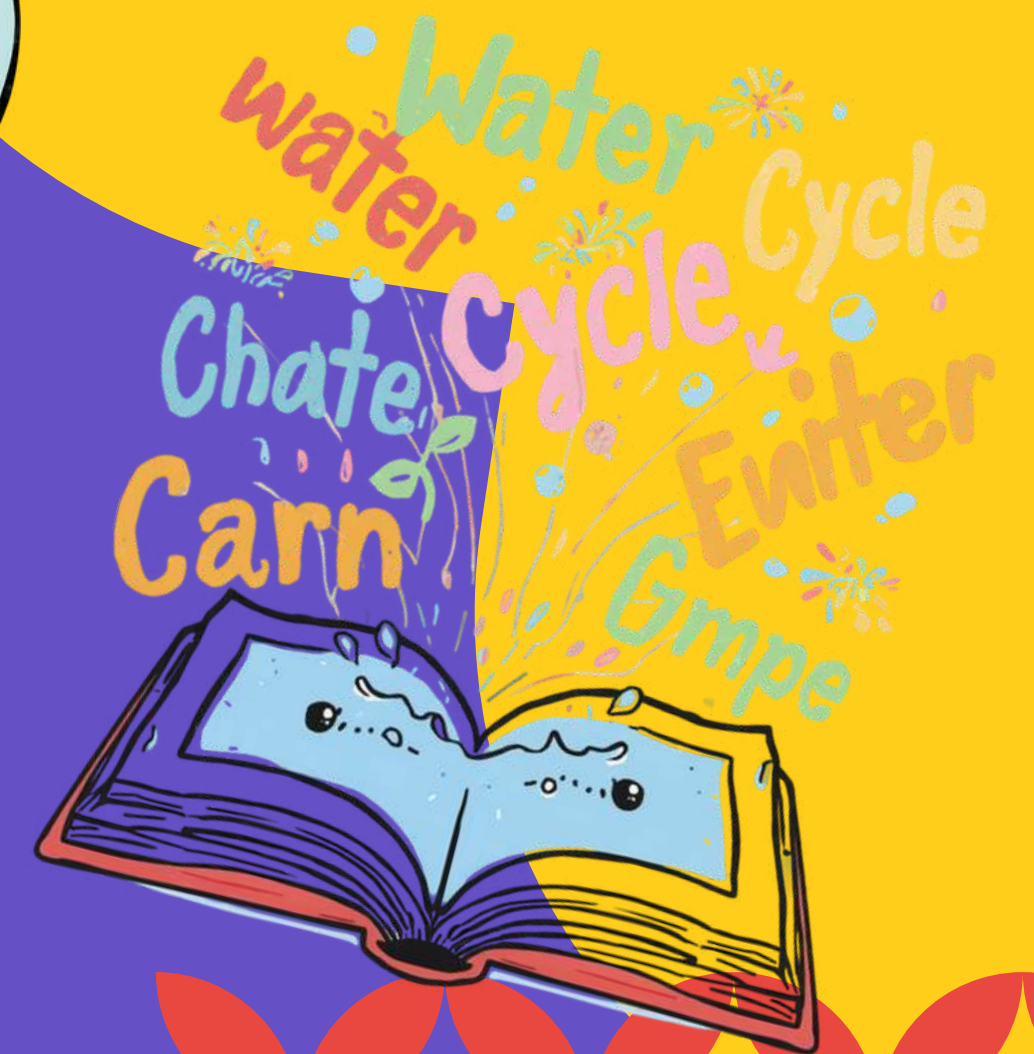
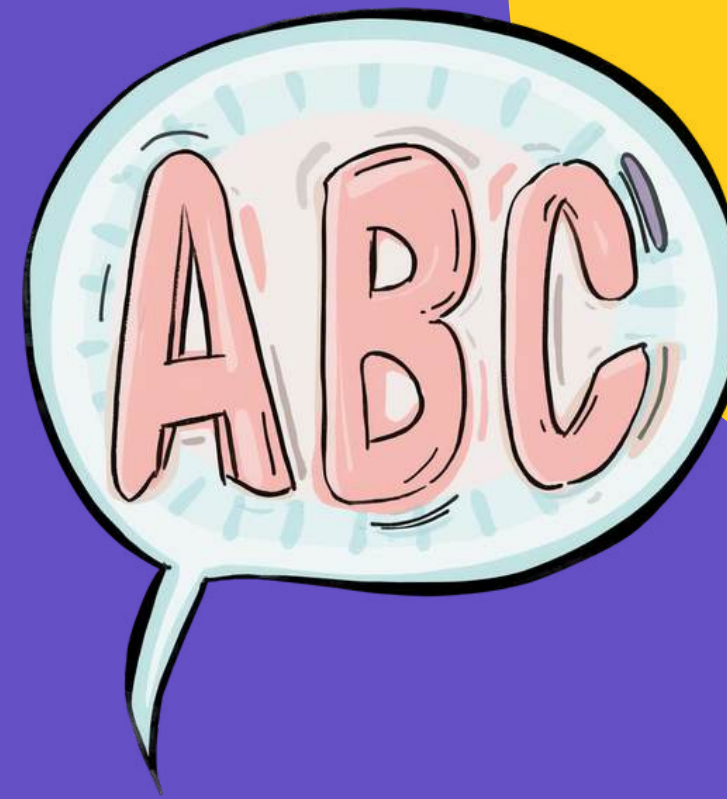
Evaporation

Condensation

Precipitation / Rain

Water Vapour

Cloud • Sun / Heat • Droplet • Cycle • Observe — You learned 10 new science words today!



Remember This!

Water never disappears — just changes form!

Water cycle happens every day, all year round

Same water goes round and round — you are a real water scientist now!

Take-home Task: Leave your bag on the window overnight and draw what you see tomorrow morning! Keep exploring the amazing water cycle!

